

NTO1064 STUDENT PROJECT

Boolean Expression & Truth Table Simulator

0.1 Boolean Expression & Truth Table Simulator

0.1.1 Overview

In this project, you will create an interactive Boolean Expression Simulator using Object-Oriented Programming (OOP) concepts in C++. The simulator will accept a Boolean expression containing a maximum of three variables (A, B, C) and up to three Boolean operations. It will provide detailed explanations for each operator, evaluate the expression, and display a clear truth table showing all possible input-output combinations.

0.2 Project Objectives

- Deepen your understanding of Object-Oriented Programming (classes, inheritance, polymorphism, encapsulation).
- Enhance your ability to parse and evaluate Boolean expressions programmatically.
- Implement robust algorithms for generating truth tables.
- Provide clear, user-friendly interactions through a console-based interface.
- Practice file handling in C++ (saving and loading data).

0.3 Functional Requirements

Your simulator must perform the following:

1. Input Boolean Expressions:

- Accept user input of Boolean expressions involving variables (A, B, C) and operators (AND, OR, NOT, XOR, NAND, NOR).
- Limit expressions to a maximum of three operations.

2. Operator Explanations:

- Automatically detect and clearly explain each operator used in the expression, including examples of expected outputs.

3. Truth Table Generation:

- Generate and display a complete truth table for the entered Boolean expression, covering all possible combinations of inputs.

4. File Saving and Loading:

- Save the expression details, operator explanations, and the truth table into a text file.
- Provide the ability to load previously saved expressions and regenerate their truth tables.

0.4 OOP Structure Guidelines

- **Abstract Base Class:**

- Boolean Operator: defines a common interface for Boolean operators.

- **Derived Classes:**

- Specific implementations such as AND Operator, OR Operator, NOT Operator, etc., inheriting from the abstract base class.

- **Core Classes:**

- Boolean Expression: Handles parsing, evaluation, and expression management.
- Truth Table: Manages the generation and presentation of truth tables.

0.5 Assessment Breakdown (30% of your final grade)

Criterion	Marks (%)
OOP Design & Implementation	25%
Encapsulation & Data Protection	10%
Inheritance & Polymorphism	15%
Parsing & Expression Evaluation	20%
Truth Table Generation	15%
File Handling (Saving/Loading)	10%
User Interaction & Usability	5%

0.6 Example User Interaction

For a demonstration of the Boolean Expression Simulator, watch the following video:

[Click Here to Watch Sample Solution](#)

```
*** BOOLEAN TRUTH TABLE SIMULATOR ***

Enter Boolean Expression (max 3 operators, variables A, B, C):
(A AND B) OR (NOT C)

Operators Detected and Explained:
- AND: True only if both inputs are true.
- OR: True if at least one input is true.
- NOT: Inverts the input.

Generating Truth Table...
| A | B | C | NOT C | A AND B | (A AND B) OR (NOT C) |
|---|---|---|-----|-----|-----|
| 0 | 0 | 0 | 1      | 0        | 1          |
| 0 | 0 | 1 | 0      | 0        | 0          |
| 0 | 1 | 0 | 1      | 0        | 1          |
| 0 | 1 | 1 | 0      | 0        | 0          |
| 1 | 0 | 0 | 1      | 0        | 1          |
| 1 | 0 | 1 | 0      | 0        | 0          |
| 1 | 1 | 0 | 1      | 1        | 1          |
| 1 | 1 | 1 | 0      | 1        | 1          |

Would you like to save this expression and truth table to a file?
(Y/N): Y
Enter filename: truthtable-example.txt

Expression and truth table saved successfully!
```

0.7 Submission

Submit your well-documented and structured C++ source code files, along with example saved truth tables. ALL CODE FILES SHOULD BE ZIPPED, AND THE CORRESPONDING DOCUMENTATIONS SHOULD BE PUT IN A SINGLE FOLDER AND ZIPPED SEPARATELY. Failure to follow these instructions will result in penalties.

Good luck and have fun exploring Boolean logic with OOP in C++!